



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Michael Bonacci
June 5, 2024



Executive summary

Scope

Analyzed historical Falcon 9 launch records with SQL, Python visualizations, geographic analysis and four classification models.

Verified model result

The saved notebook reports 88.89% test accuracy for the Decision Tree and 83.33% for each of the other three models.

Interpretation

The Decision Tree led by one correct prediction on an 18-launch test set. Results are educational and require stronger validation before operational use.

Research question and business context

Research question

Can launch characteristics predict the first-stage landing outcome recorded in the course dataset?

Business relevance

Recovery and reuse can affect launch economics. Landing predictions could support analysis of reuse opportunities.

Boundary

The project predicts a course-defined landing class. It does not estimate actual launch costs, future reuse or competitive bids.

Dataset scope

Analysis	Records	Coverage / structure
EDA and modeling	90	2010-06-04 to 2020-11-05
Encoded model features	90	83 input columns
SQL queries	101	2010-06-04 to 2020-12-06
Maps	56	2010-06-04 to 2018-06-04
Dashboard	56	Same flight numbers and classes as map data

The sections use different historical snapshots. Four site labels in the SQL and map data should not be confused with the three labels in the modeling data.

Data collection and preparation

Source workflow

The course describes SpaceX REST API requests and Wikipedia table scraping with BeautifulSoup. The linked analysis notebooks load supplied course CSV snapshots.

Preparation

Used pandas to inspect records and missing values. The modeling table contains 26 missing LandingPad values out of 90 rows (28.89%).

Features

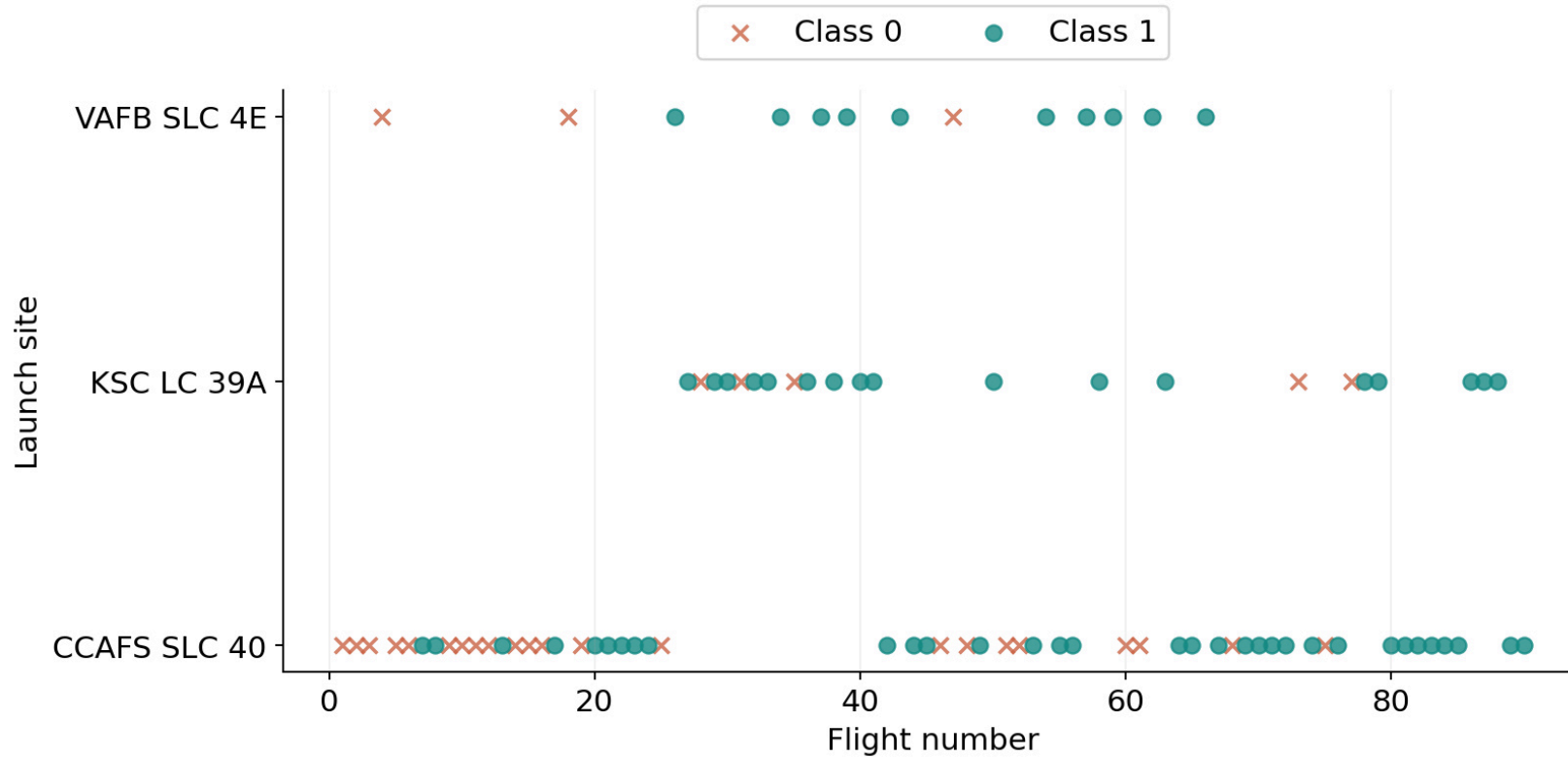
The encoded model table has 83 columns, including payload, flight information and one-hot encoded site, orbit and booster attributes.

Landing labels and class balance

Outcome group	Records	Class
True ASDS (drone ship)	41	1
True RTLS (return to launch site)	14	1
True Ocean	5	1
False / None outcomes	30	0

Class 1 totals 60 of 90 launches (66.67%). Five are ocean outcomes. Class 0 also includes no-attempt or unknown outcomes, so the label is broader than attempted landing success or recoverability.

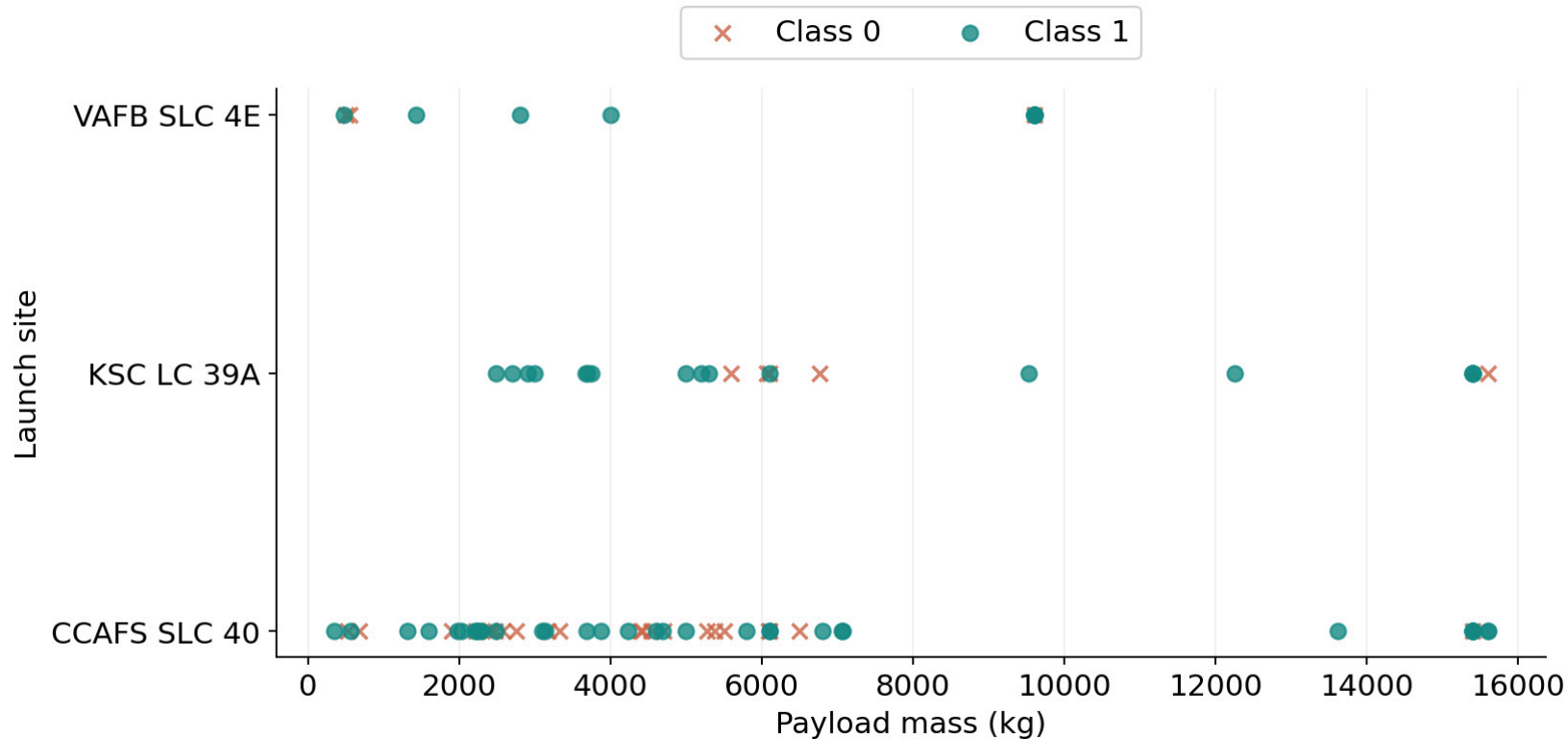
Flight number and launch site



CCAFS SLC 40 accounts for 55 of 90 records.

Class 1 shares:
CCAFS: 33/55 (60.0%)
KSC: 17/22 (77.3%)
VAFB: 10/13 (76.9%)

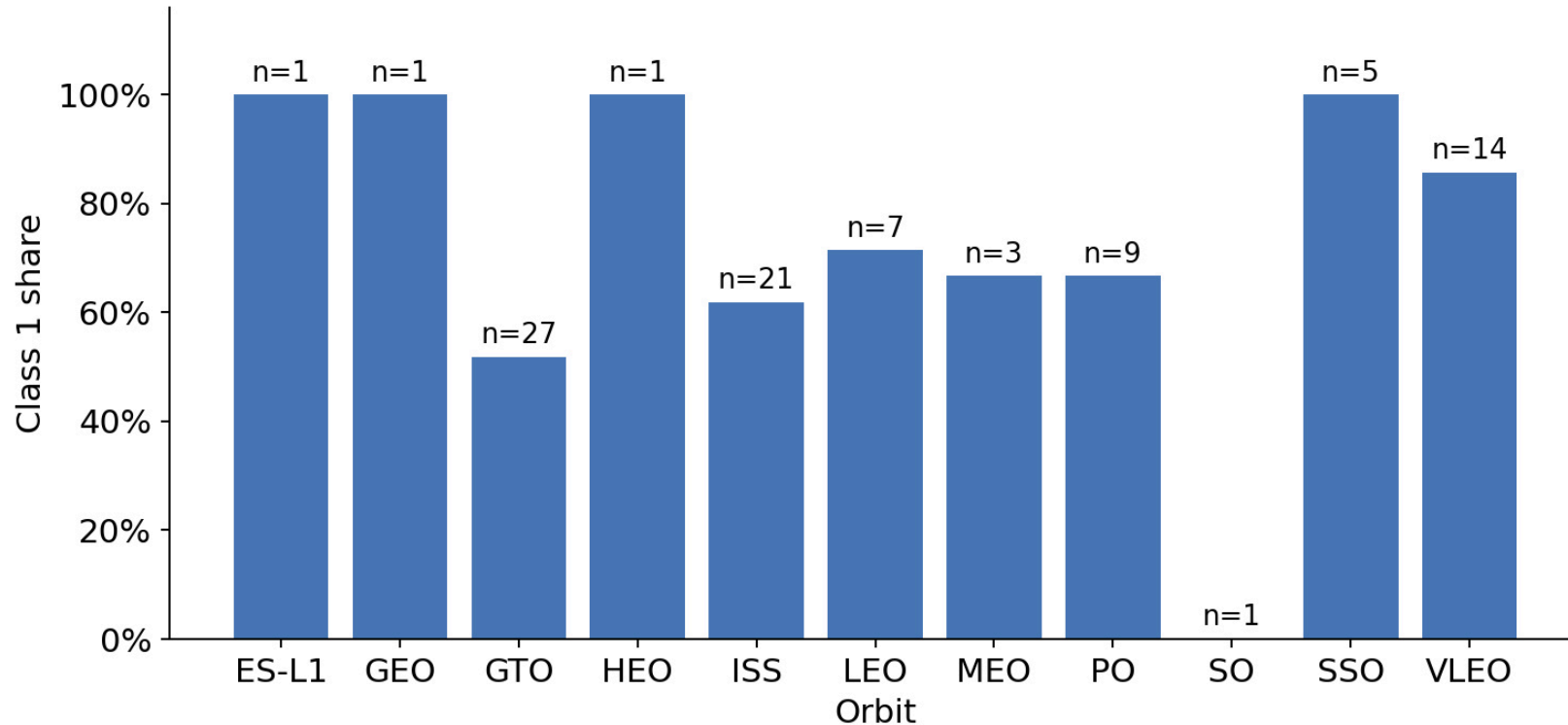
Payload mass and launch site



Successful and unsuccessful labels overlap in payload mass.

Payload alone does not cleanly separate the classes. Site comparisons also reflect different launch histories.

Landing-class share by orbit



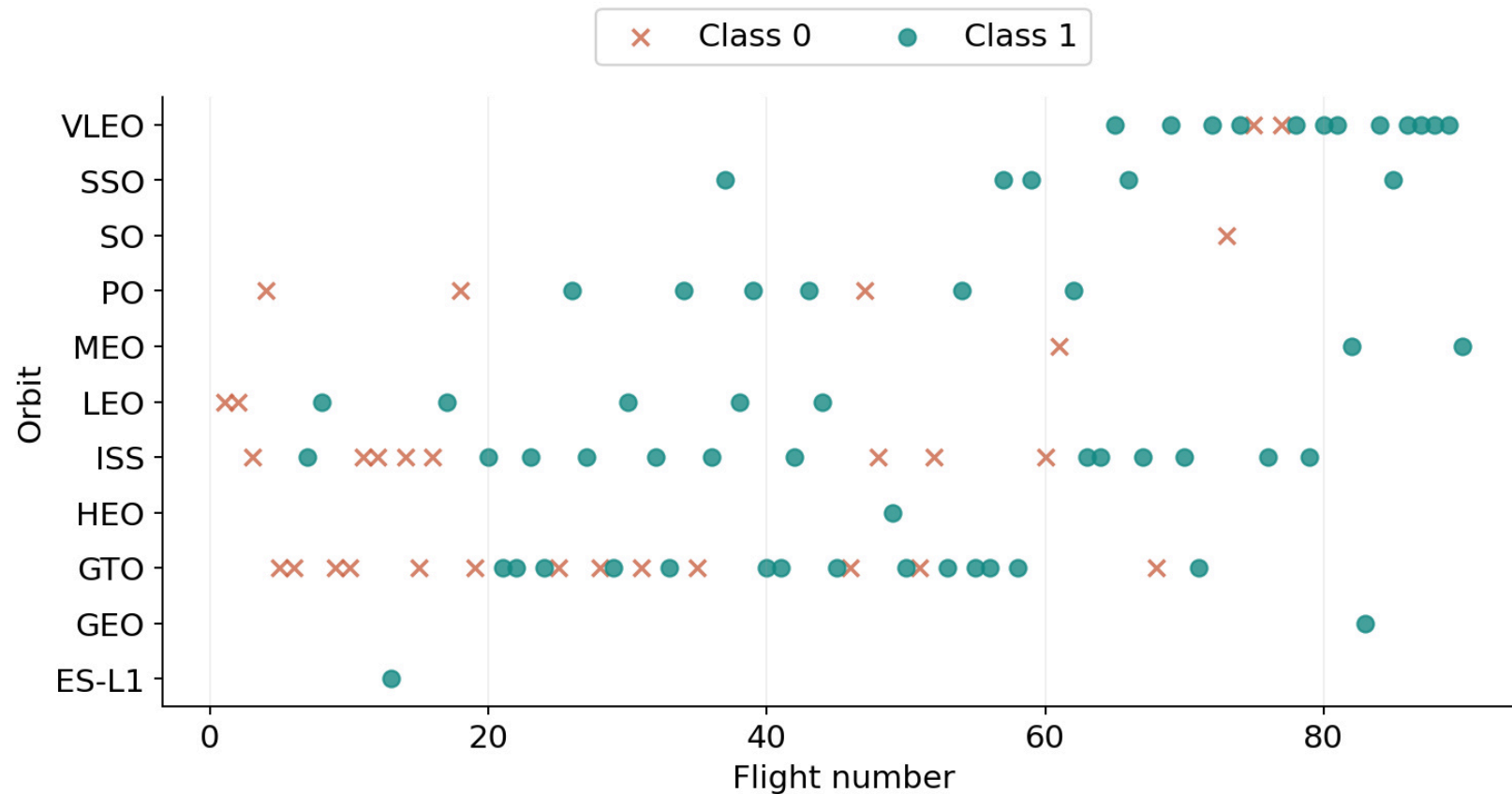
GTO: 14/27 (51.9%)

ISS: 13/21 (61.9%)

VLEO: 12/14 (85.7%)

Several 100% categories
contain only one observation.
Sample size matters.

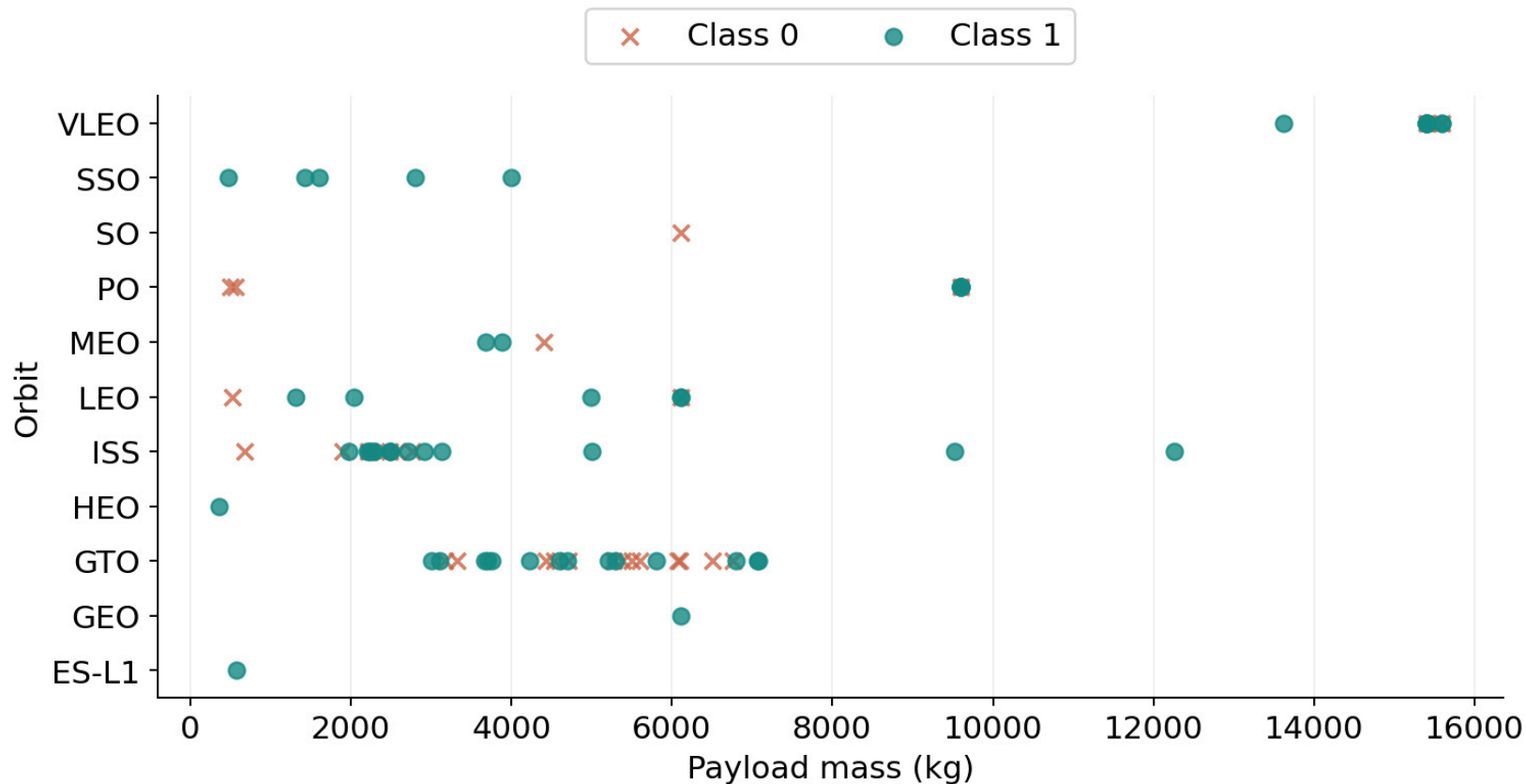
Flight number and orbit type



Orbit mix changes across the flight sequence.

Flight number also represents time and evolving technology. This complicates causal interpretation.

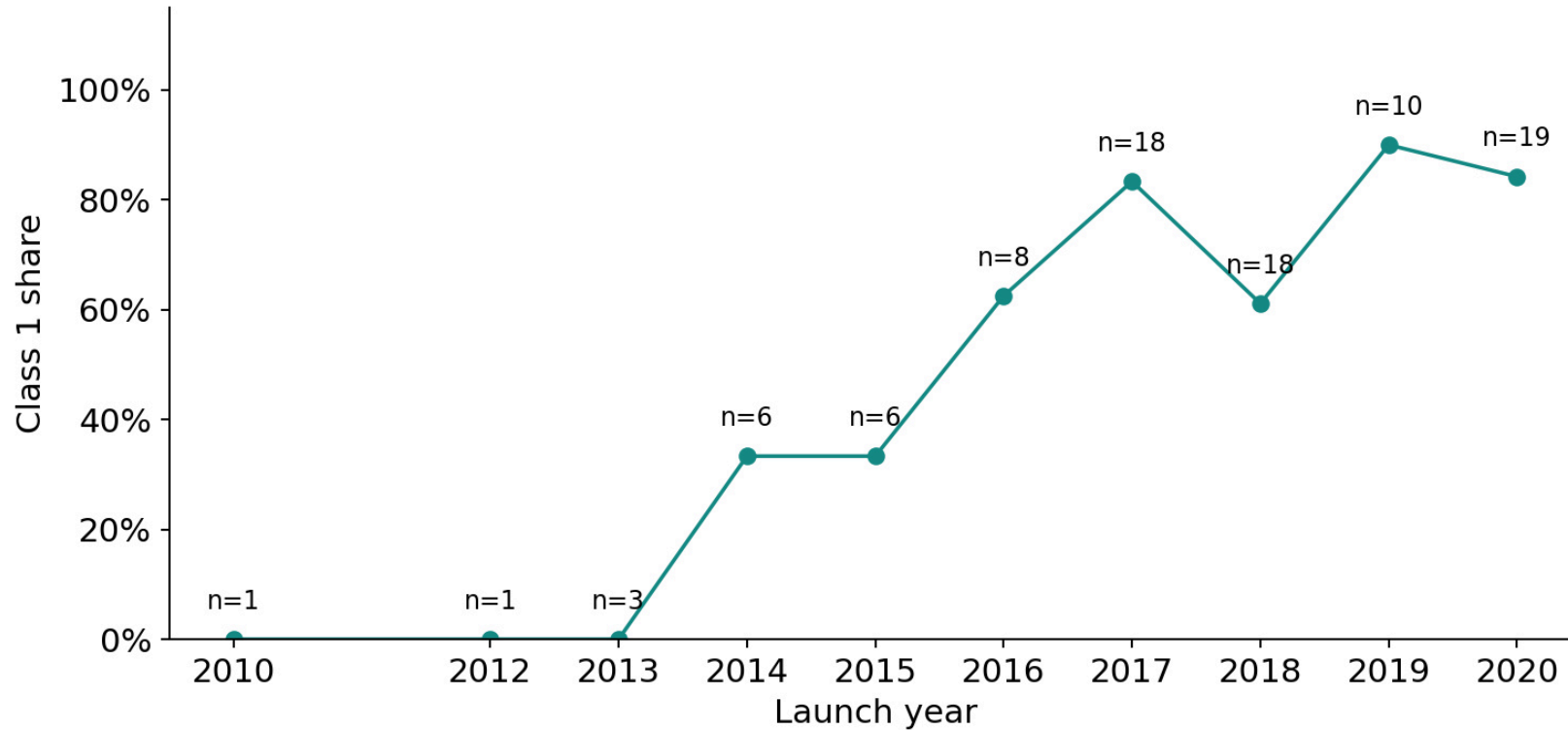
Payload mass and orbit type



Payload ranges vary by orbit.

The largest payloads concentrate in VLEO in this sample. Comparisons should account for orbit and launch era.

Yearly landing-class trend



Class 1 share rises overall but is not monotonic.

2017: 83.3%

2018: 61.1%

2019: 90.0%

2020: 84.2%

2020 ends on Nov. 5.

SQL 1: Unique launch-site labels

Site

CCAFS LC-40

CCAFS SLC-40

KSC LC-39A

VAFB SLC-4E

These are four distinct labels as stored in the SQL dataset. The modeling dataset uses a different site-label scheme.

SQL 2: First five CCA records

Date	Booster	Site
2010-06-04	F9 v1.0 B0003	CCAFS LC-40
2010-12-08	F9 v1.0 B0004	CCAFS LC-40
2012-05-22	F9 v1.0 B0005	CCAFS LC-40
2012-10-08	F9 v1.0 B0006	CCAFS LC-40
2013-03-01	F9 v1.0 B0007	CCAFS LC-40

Rows are ordered by date for a reproducible first-five result.

SQL 3: NASA CRS total payload

Payload kg

45596

The exact NASA (CRS) customer match totals 45,596 kg. This does not include every record whose customer text mentions NASA.

SQL 4: F9 v1.1 mean payload

Average kg	Launches
2,928.40	5

Exact label 'F9 v1.1': 2,928.4 kg across 5 rows. Including serial-number variants with LIKE 'F9 v1.1%' gives 2,534.67 kg across 15 rows.

SQL 5: First successful ground landing

Date

2015-12-22

The earliest recorded successful ground-pad landing is December 22, 2015.

SQL 6: Drone-ship successes, 4000 to 6000 kg

Date	Booster	Payload kg
2016-05-06	F9 FT B1022	4696
2016-08-14	F9 FT B1026	4600
2017-03-30	F9 FT B1021.2	5300
2017-10-11	F9 FT B1031.2	5200

Four records meet both strict bounds: greater than 4,000 kg and less than 6,000 kg.

SQL 7: Mission outcomes

Outcome	Count
Success	99
Success (payload status unclear)	1
Failure (in flight)	1

TRIM merges the trailing-space Success label. Keep the one payload-status-unclear record distinct. Mission outcome is different from landing outcome.

SQL 8: Maximum payload boosters

Date	Booster	Payload kg
2019-11-11	F9 B5 B1048.4	15600
2020-01-07	F9 B5 B1049.4	15600
2020-01-29	F9 B5 B1051.3	15600
2020-02-17	F9 B5 B1056.4	15600
2020-03-18	F9 B5 B1048.5	15600
2020-04-22	F9 B5 B1051.4	15600

All 12 listed launches carried the dataset maximum of 15,600 kg.

SQL 8: Maximum payload boosters (continued)

Date	Booster	Payload kg
2020-06-04	F9 B5 B1049.5	15600
2020-09-03	F9 B5 B1060.2	15600
2020-10-06	F9 B5 B1058.3	15600
2020-10-18	F9 B5 B1051.6	15600
2020-10-24	F9 B5 B1060.3	15600
2020-11-25	F9 B5 B1049.7	15600

All 12 listed launches carried the dataset maximum of 15,600 kg.

SQL 9: 2015 drone-ship failures

Date	Month	Booster	Site
2015-01-10	January	F9 v1.1 B1012	CCAFS LC-40
2015-04-14	April	F9 v1.1 B1015	CCAFS LC-40

January and April 2015 each contain one failed drone-ship landing in this dataset.

SQL 10: Landing outcomes through March 2017

Outcome	Count
No attempt	10
Failure (drone ship)	5
Success (drone ship)	5
Controlled (ocean)	3
Success (ground pad)	3
Failure (parachute)	2
Uncontrolled (ocean)	2
Precluded (drone ship)	1

31 records fall in the inclusive date interval. Tied counts remain tied even though labels are alphabetized for display.

Launch-site locations



Site	Latitude	Longitude
CCAFS LC-40	28.56230	-80.57736
CCAFS SLC-40	28.56320	-80.57682
KSC LC-39A	28.57325	-80.64690
VAFB SLC-4E	34.63283	-120.61075

The map places the source labels in Florida and California. Coordinates and naming are dataset values, not a current geospatial survey.

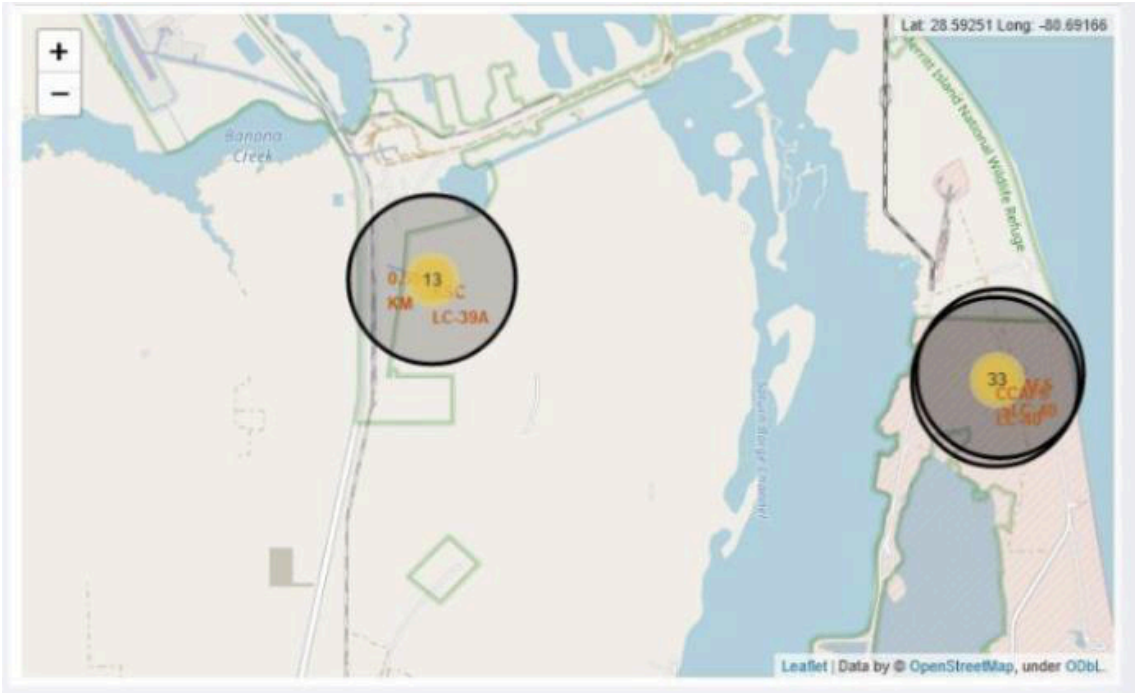
Source: [G] coordinate output and original capstone map. Names and coordinates follow the course snapshot.

Landing outcomes by mapped site

Site	Class 1	Class 0	Total	Class 1 share
CCAFS LC-40	7	19	26	26.9%
CCAFS SLC-40	3	4	7	42.9%
KSC LC-39A	10	3	13	76.9%
VAFB SLC-4E	4	6	10	40.0%

KSC LC-39A has the largest class 1 share in this 56-launch snapshot: 10 of 13 (76.9%). Raw marker overlap does not indicate success rate without counts.

Proximity analysis near CCAFS SLC-40

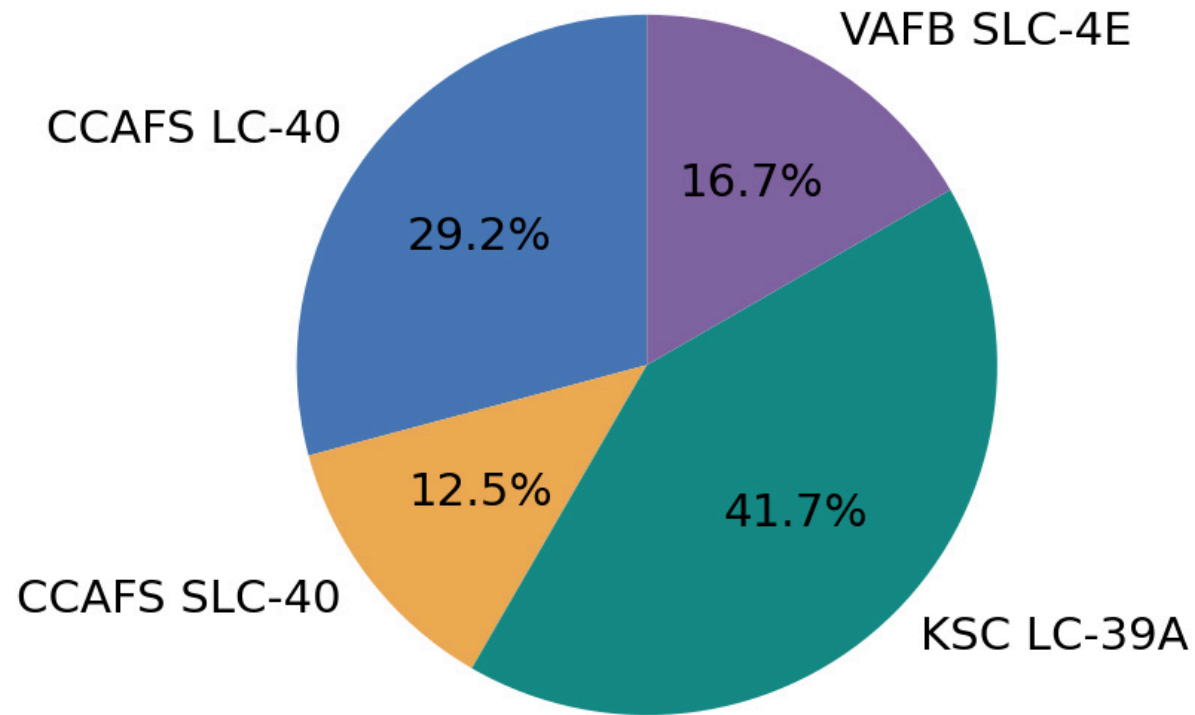


Selected feature	Distance (km)
Coastline point	0.863
Highway point	0.583
Railway point	1.285
City point	51.434

Distances measure straight-line separation to points selected in the notebook. They do not establish the nearest feature, road-travel distance or a causal effect on landing success.

Source: [G], cells 57, 71 and 72. Haversine calculation uses Earth radius 6,373 km.

Dashboard: share of successful outcomes

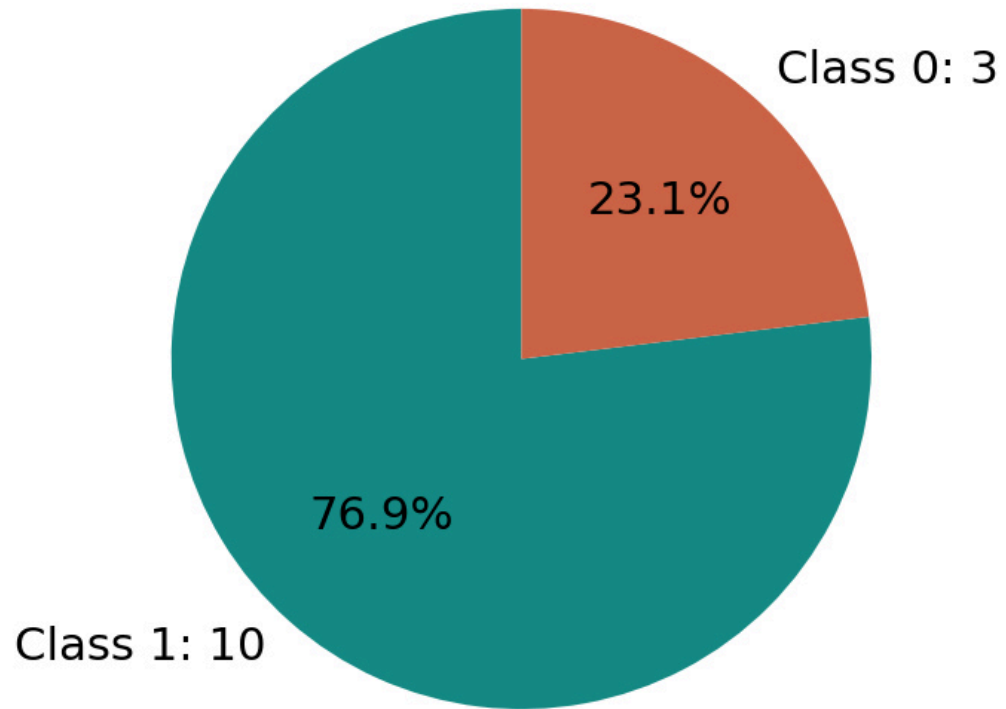


There are 24 class 1 outcomes across 56 launches.

KSC contributes 10/24 (41.7%) of those outcomes.

This pie shows share of successes, not success rate.

Dashboard: highest site success rate



KSC LC-39A:

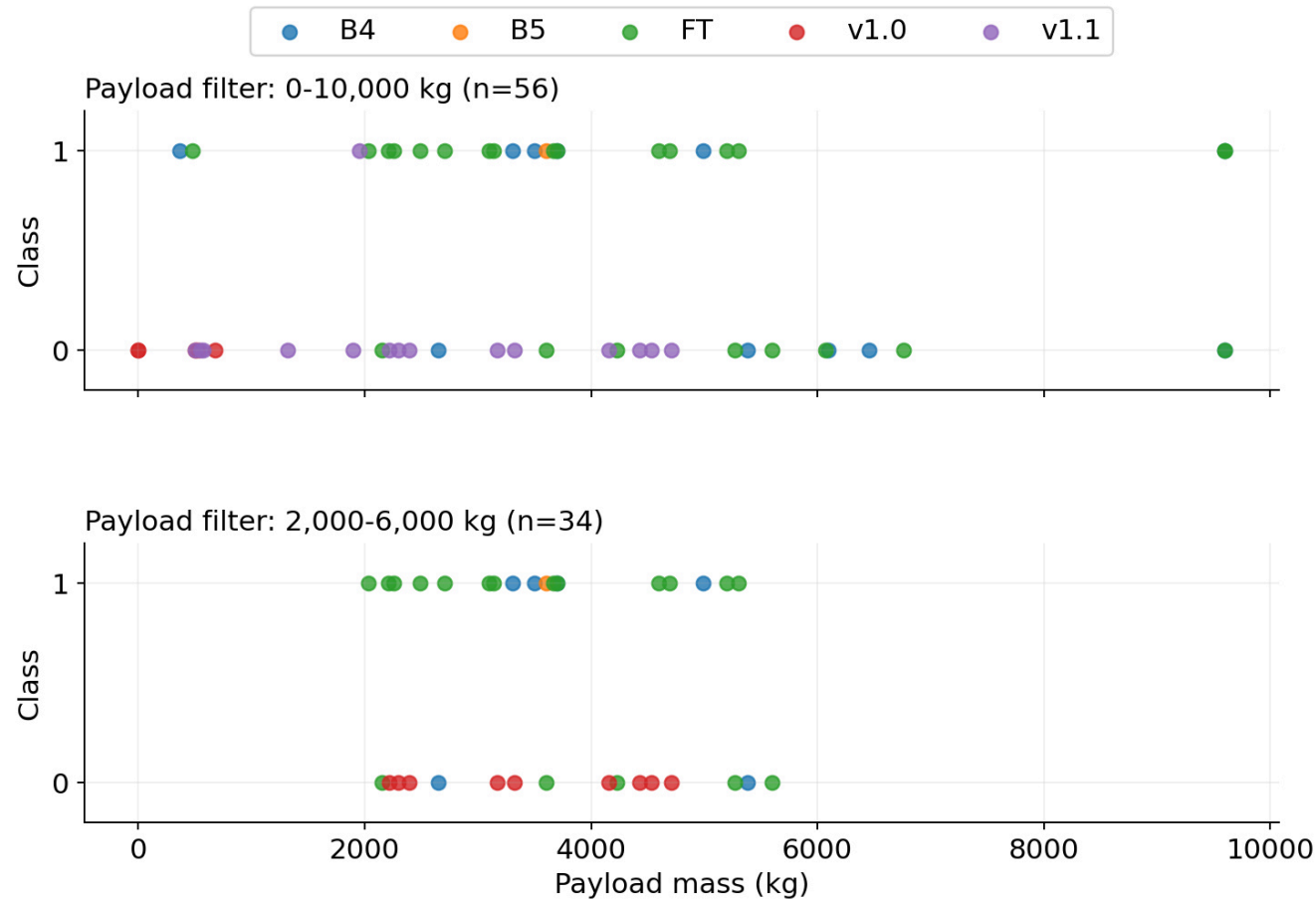
10 class 1 outcomes

3 class 0 outcomes

Rate: $10/13 = 76.9\%$

This is different from its
41.7% share of all successes.

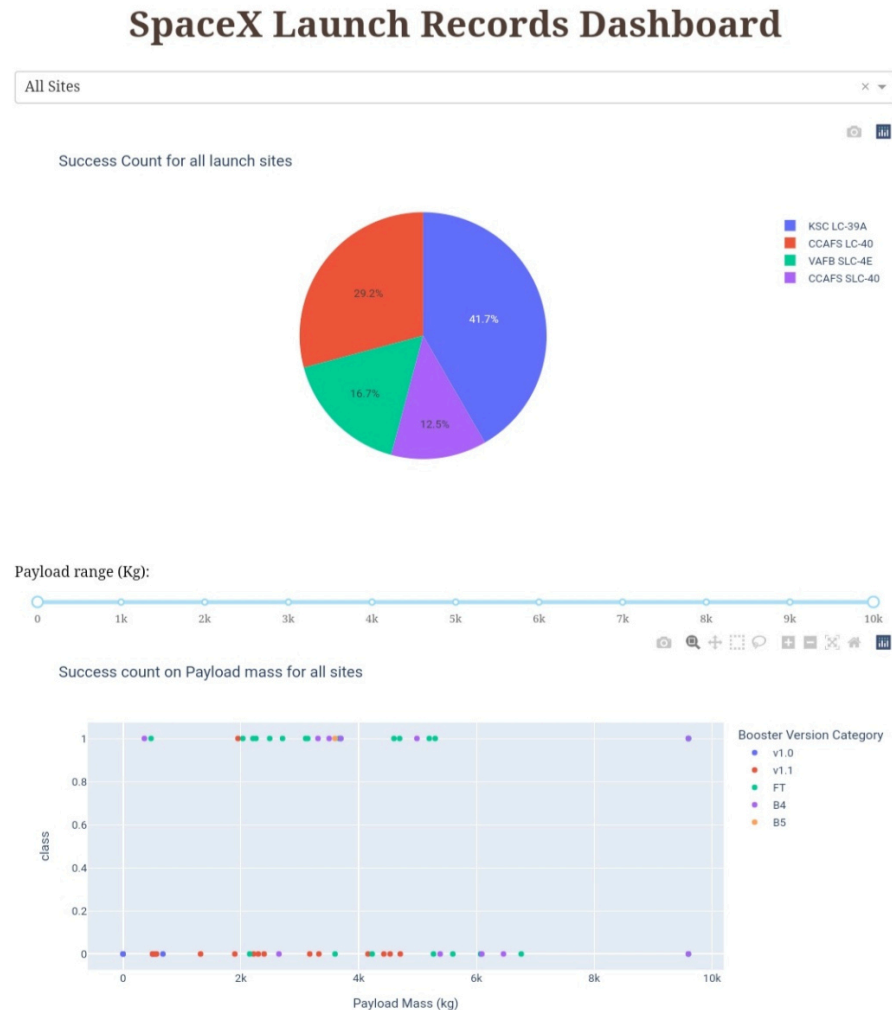
Dashboard: payload-range comparison



All-site payload filters retain different launch subsets.

Booster categories and outcomes overlap. Payload alone does not separate success from failure.

Dashboard interface evidence



The saved dashboard screenshot shows an all-sites selector, a pie chart, a payload slider and a scatterplot.

The preceding pages rebuild the required comparisons from the source CSV so the labels and counts are readable.

The repository contains screenshots, but no Dash application source in that folder.

Modeling and evaluation setup

Features and split

90 rows and 83 encoded features. The notebook uses 72 training observations and 18 test observations with `random_state=2`.

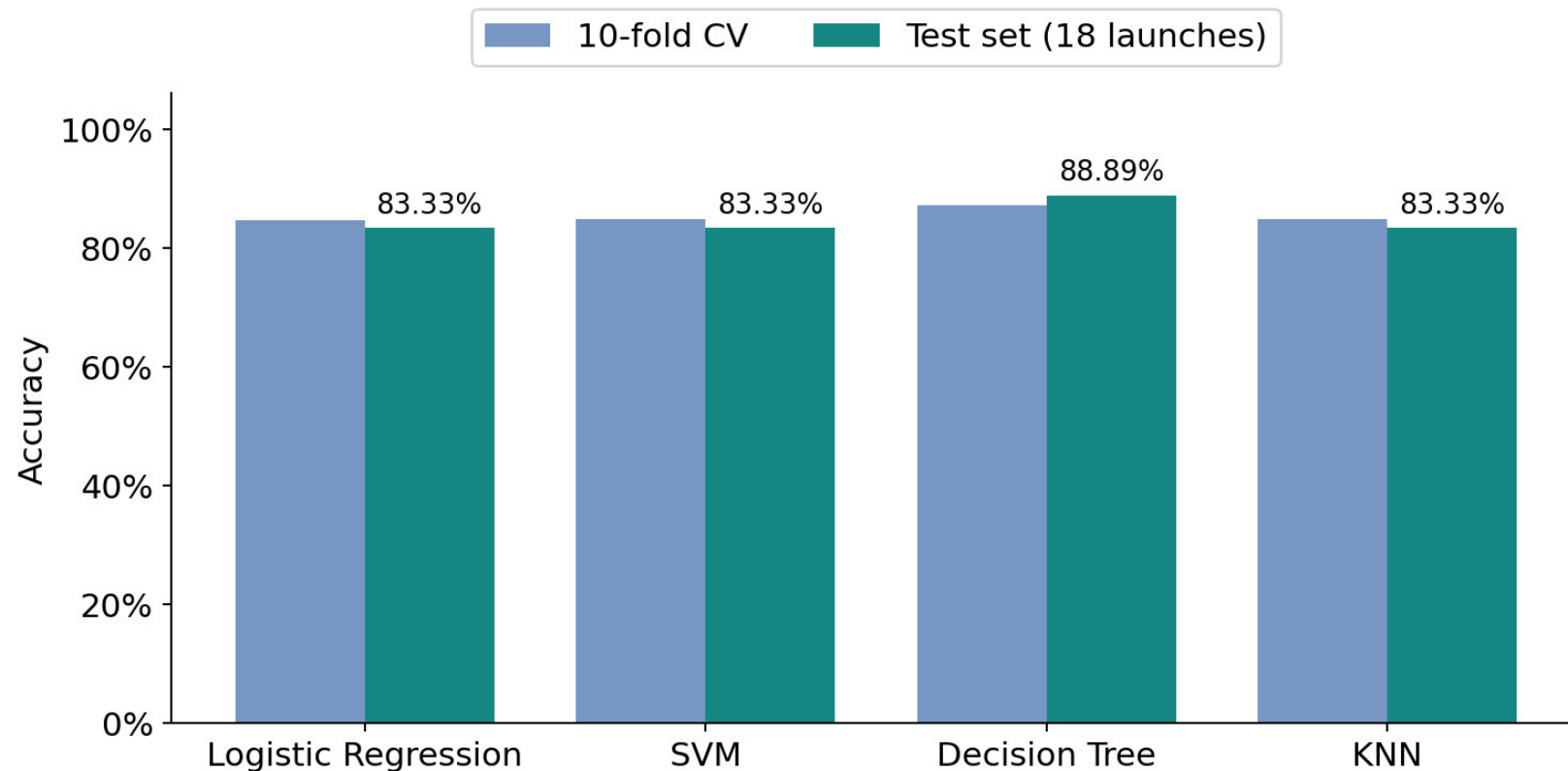
Model selection

GridSearchCV uses 10-fold cross-validation for Logistic Regression, SVM, Decision Tree and KNN.

Historical results

The following scores and confusion matrices come from saved notebook outputs. They describe the original run, not a newly retrained model.

Verified classification accuracy



Decision Tree: 16/18 correct (88.89%).

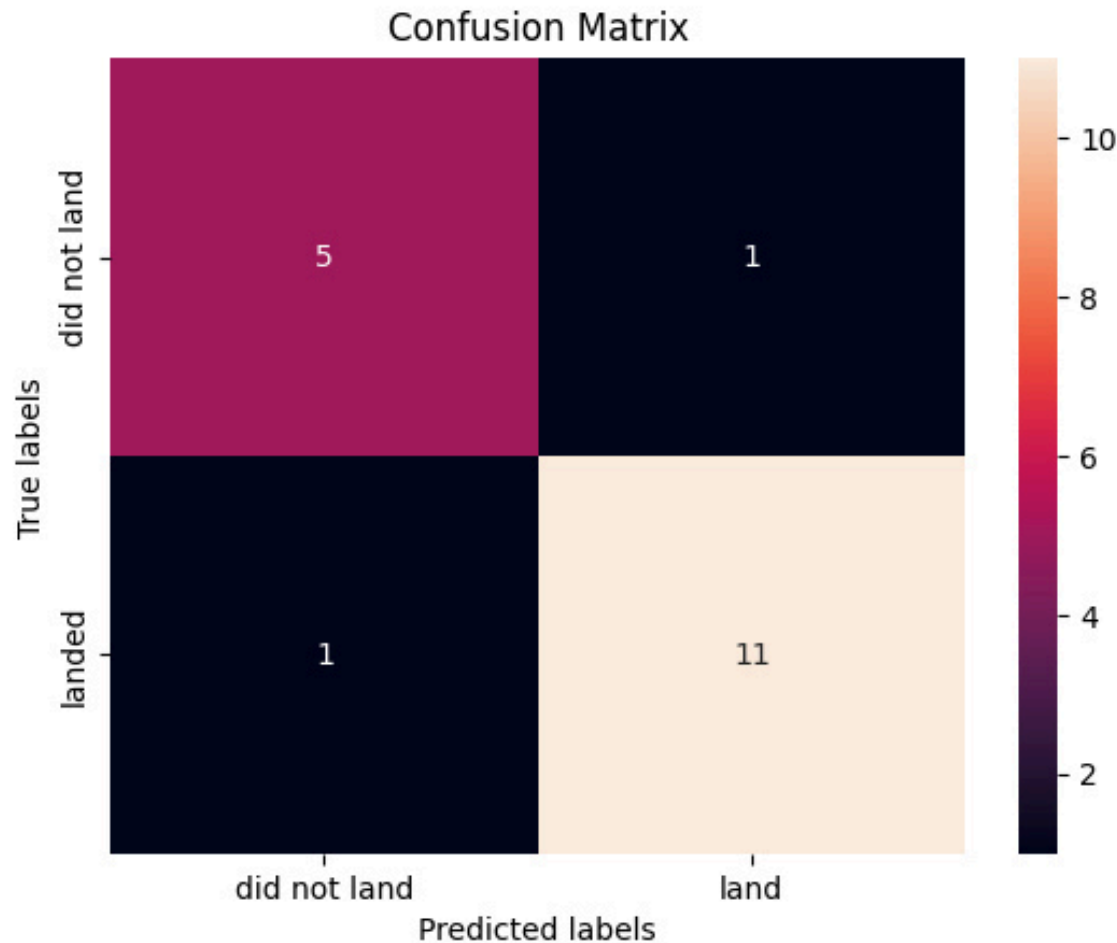
Other models: 15/18 correct (83.33%).

The original 94% slide claim does not match the saved notebook.

Selected hyperparameters

Model	Selected parameters	CV accuracy
Logistic Regression	C=0.01; L2; lbfgs	84.64%
SVM	C=1; gamma=0.0316228; sigmoid	84.82%
Decision Tree	gini; depth=10; sqrt features; leaf=4; split=5; best splitter	87.14%
KNN	k=10; p=1; algorithm=auto	84.82%

Decision Tree confusion matrix

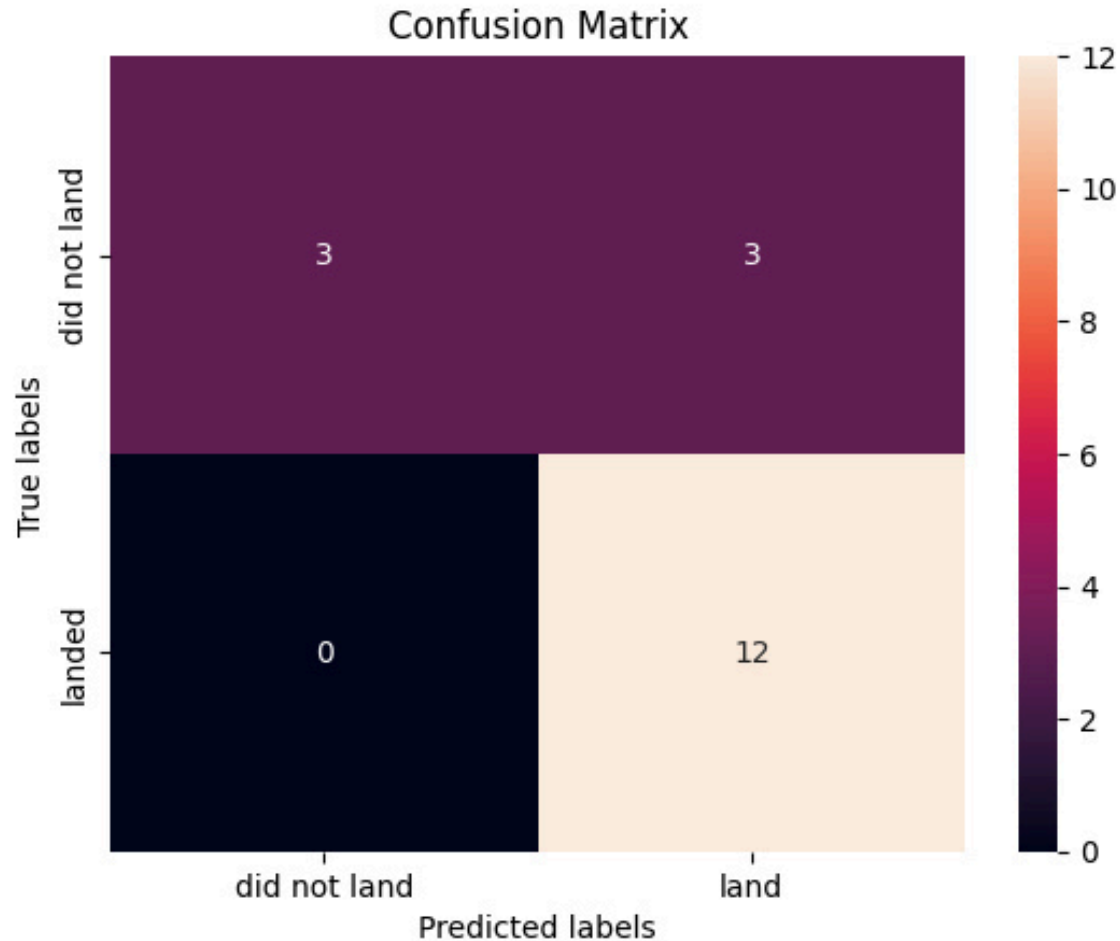


True negatives: 5
False positives: 1
False negatives: 1
True positives: 11

Accuracy: $16/18 = 88.89\%$
Precision: $11/12 = 91.67\%$
Recall: $11/12 = 91.67\%$
F1: 91.67%

Positive class = course class 1.

Other models: error comparison



Logistic Regression, SVM and KNN each show:

TN=3, FP=3, FN=0, TP=12

Accuracy: 83.33%

Precision: 80.00%

Recall: 100.00%

F1: 88.89%

The Decision Tree reduced false positives from 3 to 1, while adding one false negative.

Model limitations

Preprocessing leakage

The original notebook fits StandardScaler before the train/test split. Future evaluation should fit preprocessing inside a training-only cross-validation pipeline.

Search and reproducibility

The tree search includes an invalid `max_features="auto"` option and no fixed tree seed. Some cells label training scores as test scores. The final test table is the consistent saved result.

Small test set

One prediction changes test accuracy by 5.56 percentage points. A chronological holdout, baseline comparison and feature-availability review would strengthen validation.

Conclusions

Analytical findings

Launch characteristics, time and orbit show patterns associated with landing labels. These relationships do not establish causation.

Model comparison

The Decision Tree has the highest test accuracy in the saved run at 88.89%, with one fewer error than each alternative. This is a sample-specific result.

Business meaning

The workflow demonstrates data preparation, querying, visualization and classification. A useful launch-cost model would need cost data and a clearer recovery/reuse target.

Appendix: corrected SQL 1-2

Unique launch-site labels

```
SELECT DISTINCT TRIM(Launch_Site) AS Site  
FROM launches  
ORDER BY Site;
```

First five CCA records

```
SELECT Date, TRIM(Booster_Version) AS Booster, Launch_Site AS Site  
FROM launches  
WHERE Launch_Site LIKE 'CCA%'  
ORDER BY Date LIMIT 5;
```

Appendix: corrected SQL 3-4

NASA CRS total payload

```
SELECT SUM(PAYLOAD_MASS__KG_) AS Payload_kg  
FROM launches  
WHERE TRIM(Customer) = 'NASA (CRS)';
```

F9 v1.1 mean payload

```
SELECT AVG(PAYLOAD_MASS__KG_) AS Average_kg, COUNT(*) AS Launches  
FROM launches  
WHERE TRIM(Booster_Version) = 'F9 v1.1';
```

Appendix: corrected SQL 5-6

First successful ground landing

```
SELECT MIN(Date) AS Date
FROM launches
WHERE TRIM(Landing_Outcome) = 'Success (ground pad)';
```

Drone-ship successes, 4000 to 6000 kg

```
SELECT Date, TRIM(Booster_Version) AS Booster, PAYLOAD_MASS__KG_ AS Payload_kg
FROM launches
WHERE TRIM(Landing_Outcome) = 'Success (drone ship)'
      AND PAYLOAD_MASS__KG_ > 4000
      AND PAYLOAD_MASS__KG_ < 6000
ORDER BY Date;
```

Appendix: corrected SQL 7-8

Mission outcomes

```
SELECT TRIM(Mission_Outcome) AS Outcome, COUNT(*) AS Count
FROM launches
GROUP BY TRIM(Mission_Outcome)
ORDER BY Count DESC;
```

Maximum payload boosters

```
SELECT Date, TRIM(Booster_Version) AS Booster, PAYLOAD_MASS__KG_ AS Payload_kg
FROM launches
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_)
FROM launches)
ORDER BY Date;
```

Appendix: corrected SQL 9-10

2015 drone-ship failures

```
SELECT Date, CASE strftime('%m', Date)
  WHEN '01' THEN 'January'
  WHEN '04' THEN 'April'
  ELSE strftime('%m', Date) END AS Month, TRIM(Booster_Version) AS Booster, Launch_Site AS Site
FROM launches
WHERE TRIM(Landing_Outcome) = 'Failure (drone ship)'
  AND strftime('%Y', Date) = '2015'
ORDER BY Date;
```

Landing outcomes through March 2017

```
SELECT TRIM(Landing_Outcome) AS Outcome, COUNT(*) AS Count
FROM launches
WHERE Date BETWEEN '2010-06-04'
  AND '2017-03-20'
GROUP BY TRIM(Landing_Outcome)
ORDER BY Count DESC, Outcome;
```

Sources: project notebooks

[M] Machine learning notebook

[E] EDA visualization notebook

[S] SQL notebook

[G] Folium notebook

[W] Data-wrangling notebook

[H] Dashboard image references

Sources: data and revision scope

[D1] EDA and label data: dataset_part_2.csv

[D1] Encoded features: dataset_part_3.csv

[D2] SQL data: SpaceX.csv

[D3] Map data: spacex_launch_geo.csv

[D4] Dashboard data: spacex_launch_dash.csv

Completion work: reconstructed EDA and dashboard charts, reran corrected SQL, and reconciled saved model outputs. This revision does not claim a new model-training run or new data collection from the live SpaceX API.